

Key Challenges Faced

Main technical and design challenges during AURA implementation

01

Dataset Strategy Change

Dual-dataset fusion was replaced by one consistent dataset to reduce feature mismatch and improve reliability.

02

First-Semester GPA Gap

Some students had no official GPA, so provisional academic indicators had to be handled carefully.

03

Model Selection Trade-off

Random Forest, XGBoost and linear models were compared because accuracy alone was not enough; recall was critical for

04

Model-to-System Integration

The deployed model had to match the web input format, feature order, encoding and missing-value logic exactly.

05

Email Delivery Limits

Bulk alerts were affected by SMTP rate limits and invalid test emails, so advisor-controlled sending was required.

Design response: AURA prioritised reliable features, interpretable risk output, and advisor control over sensitive student communication.